

Sorting: evaluation, improvements, and comparisons

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Overview

- 1 Evaluation of sorting algorithms
- 2 Divide & Conquer. Improvement of merge sort: Quick sort
- 3 Improvement of of insertion sort: Shell Sort
- 4 Beyond comparison based sorting; Bucket and Radix Sort
- 5 Comparison of sorting algorithms

1 Evaluation of sorting algorithms

2 Divide & Conquer. Improvement of merge sort: Quick sort

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5 Comparison of sorting algorithms

What is sorting

■ Input

- An array A of data records
- A Key value for each data record
- A comparison function

■ Output

- Permutation of A such that
- if $i < j$, then $A[i] < A[j]$

Why sort

- Sorting algorithms are among the most frequently used algorithms in computer science

Searching : Allows binary search of an N -element array in $O(\log N)$ time

Top elements : Allows $O(1)$ time access to k -th largest element in the array for any k

Detect duplicates : Allows easy detection of any duplicates

...

Evaluation of sorting algorithms

■ Time

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- Time
- Space

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 - some algorithms are good for *partially* sorted data
 - some are good if the keys are within a certain ranges. e.g., bucket sort.

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- External: data too large to fit in memory

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- Data dependency:
 - some algorithms are good for *partially* sorted data
 - some are good if the keys are within a certain ranges. e.g., bucket sort.
- Application dependency: e.g., we only want to have the top few.
- External: data too large to fit in memory
- Parallel algorithms: data too large to run on one computer

Time: How fast is the algorithm

- The definition of a sorted array A says that for any $i < j$, $A[i] < A[j]$
- Need to at least check on each element at the very minimum, i.e., at least $O(N)$
- Could end up checking each element against every other element, which is $O(N^2)$.
- The big question is: How close to $O(N)$ can you get?

Space

How much space does the sorting algorithm require in order to sort the collection of items?

- Is copying needed?
- In-place sorting: no copying. $O(1)$ additional space
- some where in between for “temporary”, e.g. $O(\log n)$ space
- External memory sorting: data so large that does not fit in memory

In-place algorithms

An algorithm is *in-place* if it uses only a small amount of memory in addition to that needed for the original input.

Whether the following algorithms are in-place sorting?

- Merge sort?

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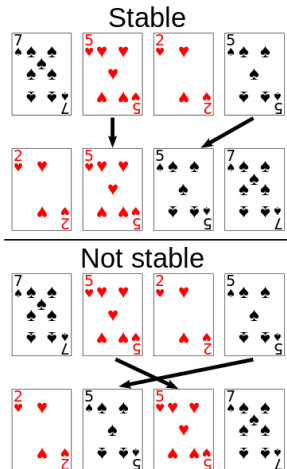
In-place algorithms

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Whether the following algorithms are in-place sorting?

- Merge sort?
- Insertion sort?

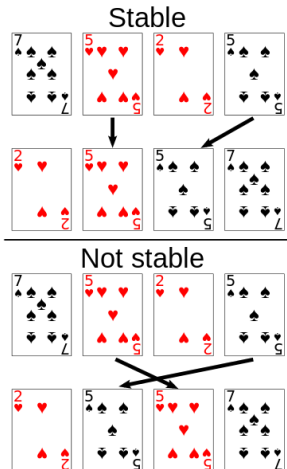
Stability



Does it rearrange the order of input data records which have the same key value (duplicates)?

- E.g. Phone book sorted by name. Now sort by county. Is the list still sorted by name within each county?
- Extremely important property for databases

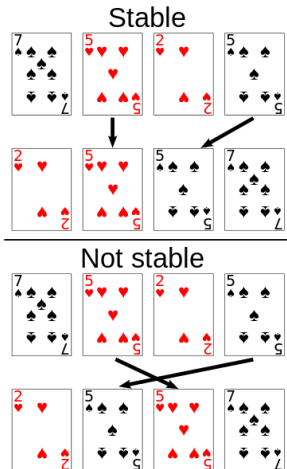
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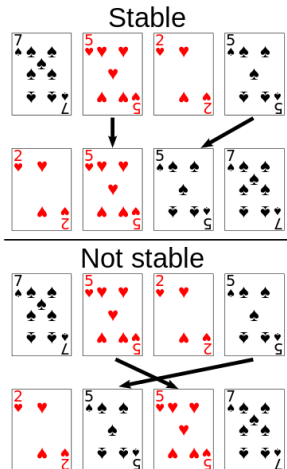
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- Whether insertion sort is stable?
- How about merge sort?

Stability



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- E.g. Phone book sorted by name. Now sort by county. Is the list still sorted by name within each county?
- Extremely important property for databases
- Whether insertion sort is stable?
- How about merge sort?
- How about heap sort?

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Quick sort

Overall idea: Divide-and Conquer

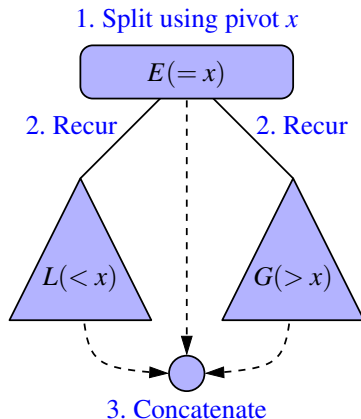
Quick-sort is a randomized sorting algorithm based on the divide-and-conquer paradigm

Divide pick a random element x (called pivot) and partition S into

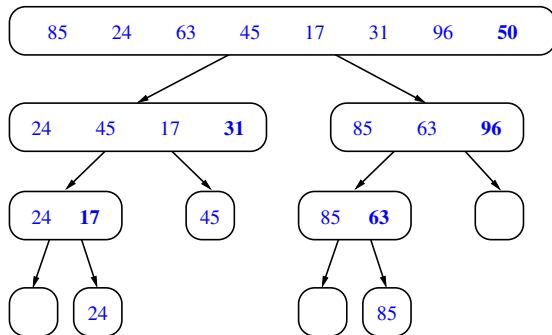
- L elements less than x
- E elements equal x
- G elements greater than x

Conquer Recursively sort L and G

Combine join L , E and G

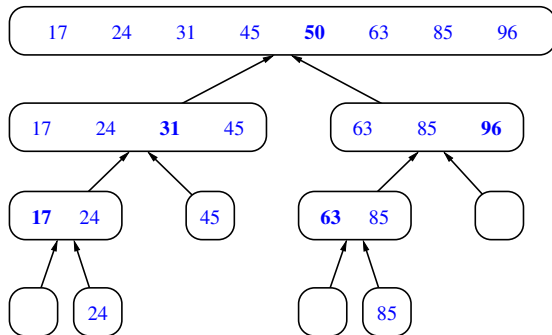


Quick sort example: divide



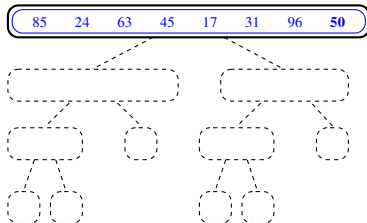
Divide process

Combine

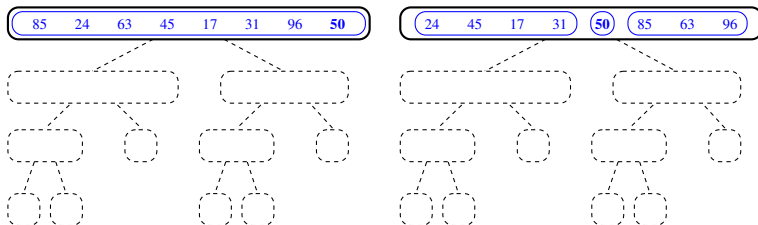


Combination process

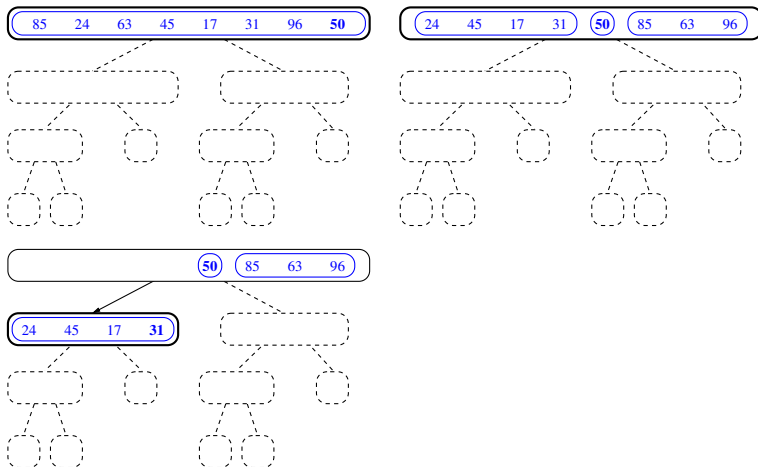
Quick sort step-by-step: recursive calls



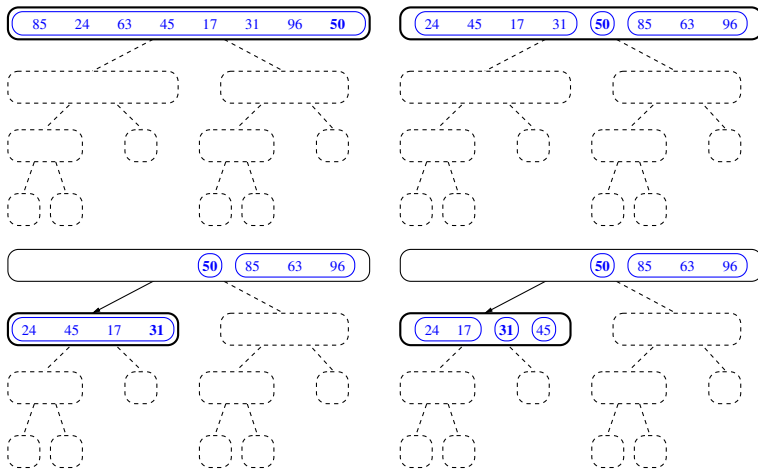
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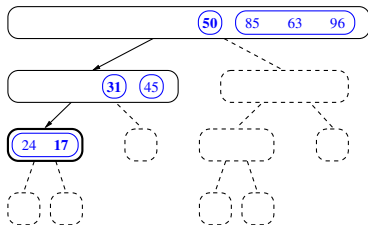
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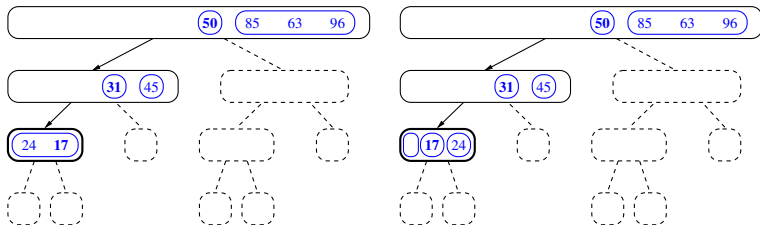
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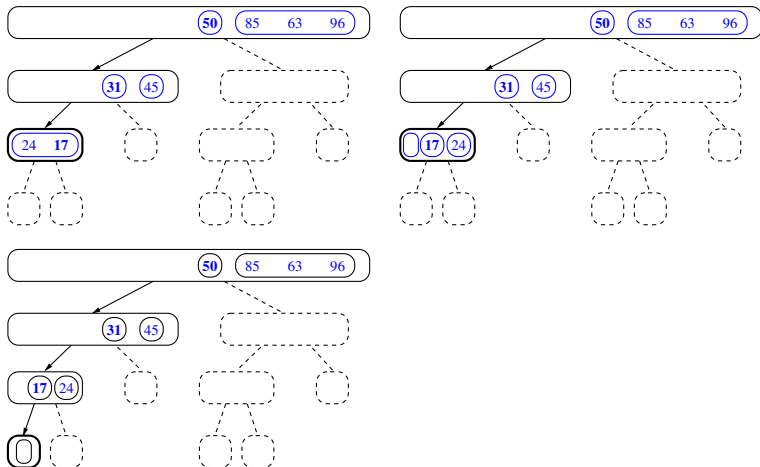
Recursive calls down to the base case



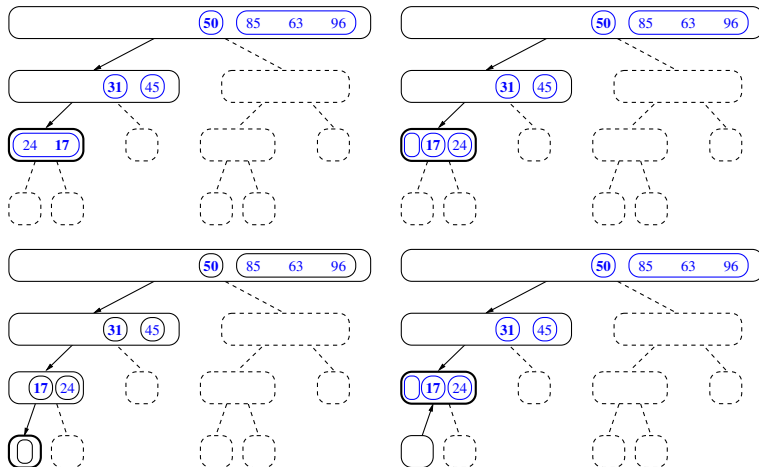
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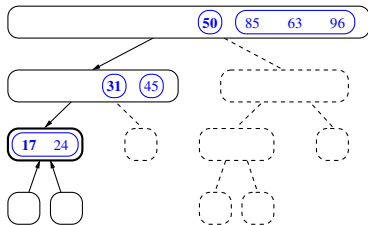
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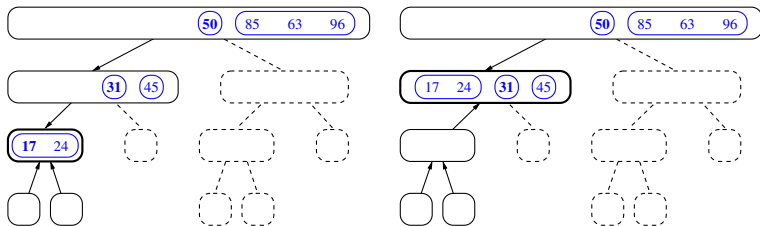
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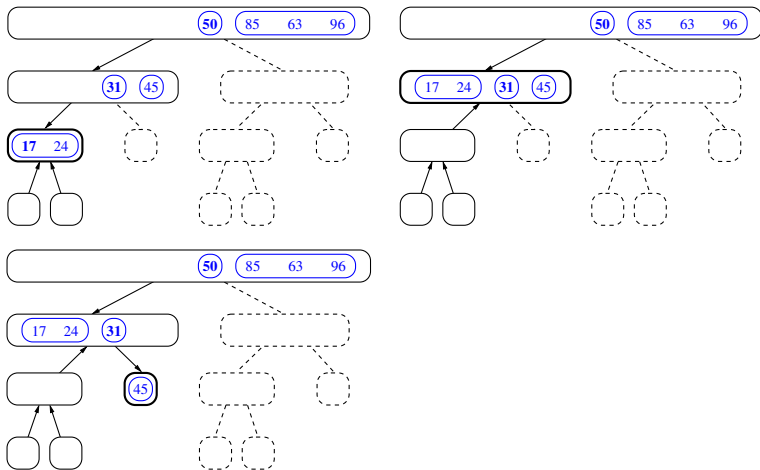
Return of recursive calls, then call (45) branch



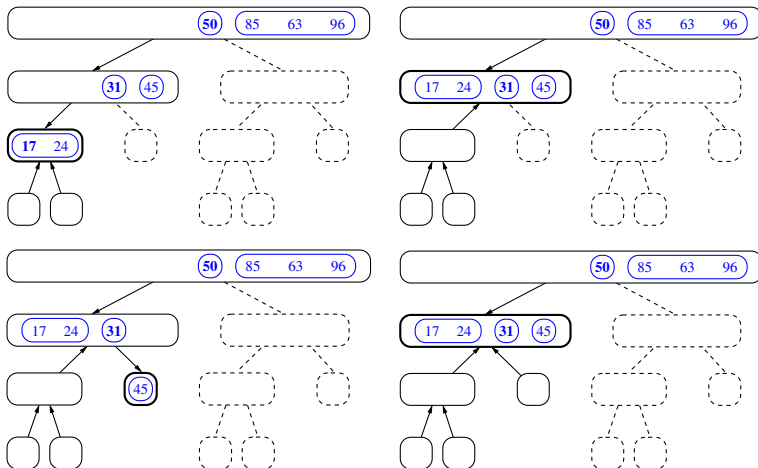
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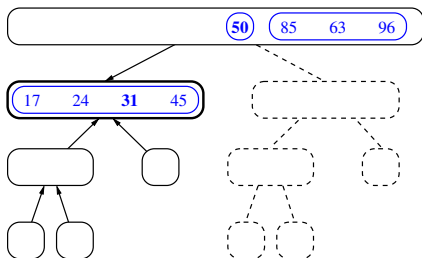
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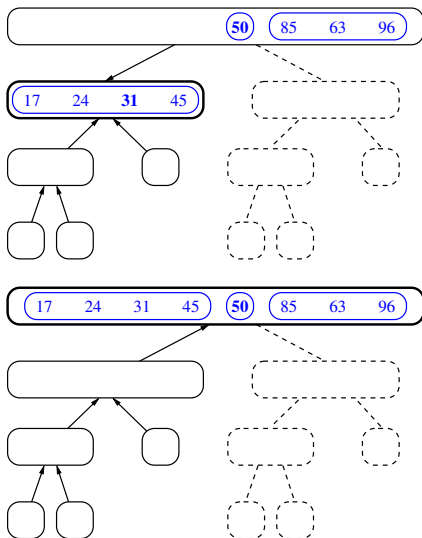
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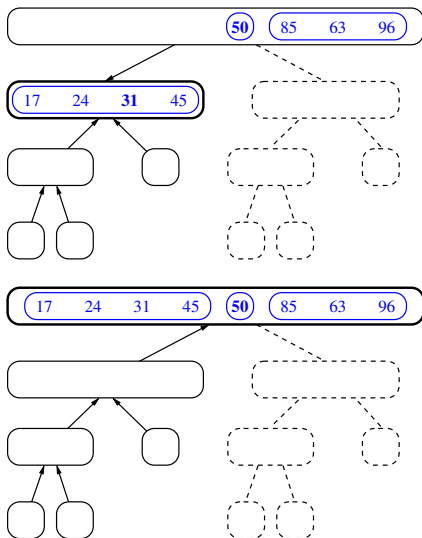
Return of the left branch, do the same for the right branch



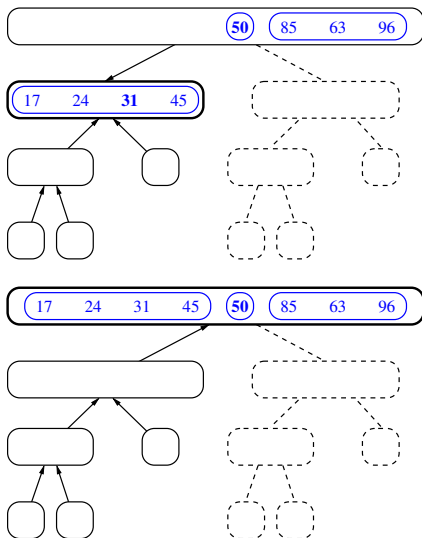
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Implementation on Queue (bad one)

Algorithm 1: quickSortBad(Queue Q)

```
4 (L, E, G)=partition(Q);  
5 quickSort(L, comp);  
6 quickSort(G, comp);
```

Implementation on Queue (bad one)

Algorithm 2: quickSortBad(Queue Q)

```
1 if  $Q.size < 2$  then
2   | return;
3 end
4 (L, E, G) = partition(Q);
5 quickSort(L, comp);
6 quickSort(G, comp);
```

Implementation on Queue (bad one)

Algorithm 3: quickSortBad(Queue Q)

```
1 if Q.size < 2 then
2   | return;
3 end
4 (L, E, G) = partition(Q);
5 quickSort(L, comp);
6 quickSort(G, comp);
7 while (!L.isEmpty()) Q.enqueue(L.dequeue());
8 while (!E.isEmpty()) Q.enqueue(E.dequeue());
9 while (!G.isEmpty()) Q.enqueue(G.dequeue());
```

Queue partition(bad one)

Algorithm 4: (L,E,G) partition(Q)

```
1 pivot = Q.first();
2 L, E, G = new LinkedQueue();
3 while Q not empty do
4     nextElement = Q.dequeue();
5     switch compare nextElement with pivot do
6         case nextElement < pivot do
7             | L.enqueue(nextElement);
8         end
9         case nextElement = pivot do
10            | E.enqueue(nextElement);
11        end
12        case nextElement > pivot do
13            | G.enqueue(nextElement);
14        end
15    end
16 end
```

Performance of partition (queue based)

- Time complexity:
 - remove each element y from Q
 - insert y into L , E or G , depending on the comparison with the pivot x
 - Each insertion and removal is at the beginning or at the end of a sequence, and hence takes $O(1)$ time
 - Thus, the partition step of quick-sort takes $O(n)$ time

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 - Thus, the partition step of quick-sort takes $O(n)$ time
- Space complexity for partition:
 - $O(n)$

Implementation on Queue (bad one) in Java

```

public static <K> void quickSort(Queue<K> S, Comparator<K> comp) {
    int n = S.size();
    if (n < 2) return;
    K pivot = S.first();
    Queue<K> L = new LinkedList<>();
    Queue<K> E = new LinkedList<>();
    Queue<K> G = new LinkedList<>();
    while (!S.isEmpty()) {
        K element = S.dequeue();
        int c = comp.compare(element, pivot);
        if (c < 0) L.enqueue(element);
        else if (c == 0) E.enqueue(element);
        else G.enqueue(element);
    }
    quickSort(L, comp);
    quickSort(G, comp);
    while (!L.isEmpty()) S.enqueue(L.dequeue());
    while (!E.isEmpty()) S.enqueue(E.dequeue());
    while (!G.isEmpty()) S.enqueue(G.dequeue());
}

```


Quick sort on Array

- A better quickSort that does not need an extra array

Algorithm 5: quickSortInPlace(*A*, int *low*, int *high*)

```
1 if low  $\geq$  high then  
2   |   return;  
3 end  
4 int p=partition(A, low, high);  
5 quickSort(A, low, p-1);  
6 quickSort(A, p+1, high);
```

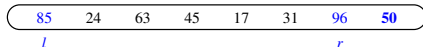
In-place Partition (rewritten from the book)

Algorithm 6: partition(A, l, r)

```

1 pivot=A[r];
2 p=r;
3 r=r-1;
4 while l ≤ r do
5     while l ≤ r and A[l] < pivot do
6         l++;
7     end
8     while l ≤ r and A[r] > pivot do
9         r--;
10    end
11    if l ≤ r then
12        swap(A[l], A[r]);
13        l++; r--;
14    end
15 end
16 swap(A[l], A[p]);
17 return l;

```



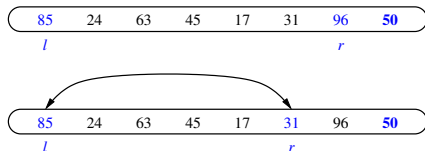
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Algorithm 7: partition(A, l, r)

```

1 pivot=A[r];
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3 r=r-1;
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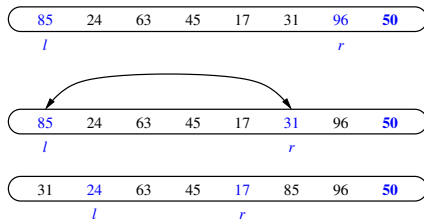
In-place Partition (rewritten from the book)

Algorithm 8: partition(A, l, r)

```

1 pivot=A[r];
2 p=r;
3 r=r-1;
4 while l ≤ r do
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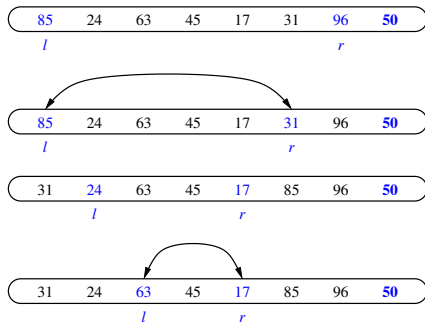
In-place Partition (rewritten from the book)

Algorithm 9: partition(A, l, r)

```

1 pivot=A[r];
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4 while l ≤ r do
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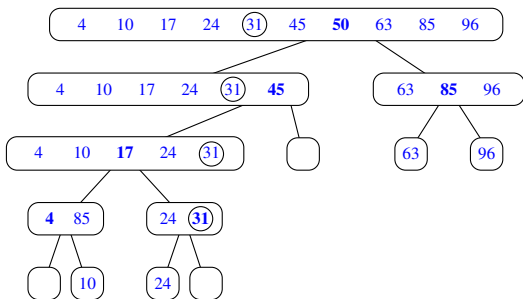
```



In-place sorting in Java

```
<K> void quickSortInPlace (K [] S, Comparator<K>comp, int a, int b){
    if (a >= b) return;
    int left = a;
    int right = b-1;
    K pivot = S[b];
    K temp;
    while (left <= right) {
        while (left<=right && comp.compare(S[left], pivot)<0) left++;
        while (left<=right && comp.compare(S[right], pivot)>0)
            right--;
        if (left <= right) {
            temp=S[left]; S[left]=S[right]; S[right]=temp;
            left++; right--;
        }
    }
    temp=S[left]; S[left]=S[b]; S[b]=temp;
    quickSortInPlace(S, comp, a, left - 1);
    quickSortInPlace(S, comp, left + 1, b);
}
```

- An execution of quick-sort is depicted by a binary tree
- Each node represents a recursive call of quick-sort and stores
- Unsorted sequence before the execution and its pivot
- Sorted sequence at the end of the execution
- The root is the initial call
- The leaves are calls on subsequences of size 0 or 1



Worst-case for Quick sort

- The worst case for quick-sort occurs when the input is sorted
 - One of L and G has size $n - 1$ and the other has size 0
- The running time is proportional to the sum of:

$$n + (n - 1) + \cdots + 2 + 1 = n(n + 1)/2 \quad (1)$$

- Thus, the worst-case running time of quick-sort is $O(n^2)$
- Can avoid this by randomize the pivot

Average-case for Quick sort

- Average case number of comparisons is $1.39N \log N$.
- Probabilistic guarantee against worst case
- Better than MergeSort because less space is needed
 - you can observe the time performance improvement when input size is very large

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Insertion Sort

Algorithm 10: Insertion-Sort(A)

```
1 for  $j = 2$  to  $A.length$  do
2    $key = A[j]$ 
3    $i = j-1$ 
4   while  $i > 0$  and  $A[i] > key$  do
5      $A[i+1] = A[i]$ 
6      $i = i-1$ 
7   end
8    $A[i+1] = key$ 
9 end
```

Best case running time is $O(n)$

- Let's focus on the number of comparisons.
- Checking number of all the 'operations' is more complicated.
- The best case is when A is already sorted. i.e.,

$$A[j-1] \leq A[j], \quad (\text{for } j = 2, 3, \dots, n)$$

- As this ensures that the condition of the *While* statement,

$$A[i] > \text{key}$$

is false for $j = 2, 3, \dots, n$,

- Inside the while loop there is only one comparison.
- The for loop checks $n-1$ elements

Thus, the number of comparison is

$$\sum_{j=2}^n 1 = n - 1.$$

Worst case number of comparisons

- In the worst case, the While condition will be tested for every possible value $i = j - 1, j - 2, \dots, 1, 0$
- The cost for each while loop changes with j
- In total, it is

$$\sum_{j=2}^{n-1} j = \frac{n(n-1)}{2} - 1.$$

Average case number of comparisons

- For an "average" list, suppose that inside each while loop, we need to do half of the comparisons (and swaps),
- Hence the total number of comparisons is

$$\sum_{j=2}^{n-1} \sum_{j=2}^n j/2 = \frac{n(n-1)}{4} - 1.$$

- So the average running time of Insertion sort is $\Theta(N^2)$
- Any sorting algorithm that only swaps adjacent elements requires $\Omega(N^2)$ time because each swap removes only one inversion.
- If we are going to do better than $O(N^2)$, we are going to have to fix more than one inversion at a time
- How can we fix more than one inversion?
- Move the elements far away with each swap

Shell sort

- Named after Donald Shell – inventor of the algorithm
- Running time is $O(N^x)$, where $x = 6/5, 5/4, 4/3, \dots$, or 2, depending on “increment sequence”
- Shell sort uses repeated insertion sorts on selected subarrays of the larger array being sorted
- Multiple passes with changing subarrays

Shell Sort Example

Input | 4 3 2 1

Shell Sort Example

Input		4	3	2	1
Two arrays with spacing=2		4	3	2	1

Shell Sort Example

Input		4	3	2	1
Two arrays with spacing=2		4	3	2	1
Sort the red array		2	3	4	1

Shell Sort Example

Input	4	3	2	1
Two arrays with spacing=2	4	3	2	1
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Sort the black array	2	1	4	3

Shell Sort Example

Input	4	3	2	1
Two arrays with spacing=2	4	3	2	1
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Sort the black array	2	1	4	3
Reduce the spacing =1	2	1	4	3
inverse 2/1	1	2	4	3

Shell Sort Example

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Shell Sort Example

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Sort the red array	2	3	4	1
Sort the black array	2	1	4	3
Reduce the spacing =1	2	1	4	3
inverse 2/1	1	2	4	3
inverse 4/3	1	2	4	3

- number of inversions in shell sort: 4
- number of inversions in insertion sort: $1+2+3=6$

Intuition of Shell Sort

- Insertion sort runs fast on nearly sorted sequences
- Immediate termination when proper spot is found
- Do several passes of Insertion sort on different subsequences of elements
- The subsequences stay sorted from pass to pass

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BucketSort (aka BinSort)

If all values to be sorted are known to be between 1 and B,

- create an array count of size B,
- increment counts while traversing the input,
- and finally output the result.
- e.g., Input=(4, 5, 1, 3, 5)
- B is 5. create 5 buckets using array:

1	
2	
3	
4	
5	

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3	
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5	

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1	
2	
3	
4	✓
5	✓

BucketSort (aka BinSort)

If all values to be sorted are known to be between 1 and B,

- create an array count of size B,
- increment counts while traversing the input,
- and finally output the result.
- e.g., Input=(4, 5, 1, 3, 5)
- B is 5. create 5 buckets using array:

1	✓
2	
3	
4	✓
5	✓

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1	✓
2	
3	✓
4	✓
5	✓

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- and finally output the result.
- e.g., Input=(4, 5, 1, 3, 5)
- B is 5. create 5 buckets using array:

1	✓
2	
3	✓
4	✓
5	✓ ✓

Complexity of Bucket Sorting

- What is the complexity of Bucket Sorting?
- Depending on B
 - What if...
 - B is a constant
 - B is proportional to n
 - B is a very large constant (2^{32})

Fixing impracticality: RadixSort

- Radix Sort: generalization of Bucket Sort for large integer keys.
- Origins go back to the 1890 census.
- Radix = “The base of a number system”
- We use 10 for convenience, but could be anything
- Ideas:
 - Bucket Sort on one digit at a time
 - After k -th sort, the last k digits are sorted
 - Set number of buckets: $B = \text{radix}$.

First pass

■ Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's										

First pass

■ Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's									478	

First pass

■ Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's								537	478	

First pass

■ Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's								537	478	9

First pass

■ Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		72 1						53 7	47 8	9

First pass

■ Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		72 1		3				53 7	47 8	9

First pass

■ Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		72 1		3				53 7	47 8	9
									3 8	

First pass

■ Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		72 1		3 12 3				53 7	47 8	9 3 8

First pass

■ Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		72 1		3 12 3				53 7 67	47 8 38	9

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		72 1		3 12 3				53 7 67	47 8 38	9

- Output of first pass: 721, 3, 123, 537, 67, 478, 38, 9
- used as the input of second pass

Second pass

■ Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's										

Second pass

■ Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's			721							

Second pass

■ Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721							

Second pass

■ Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721 123							

Second pass

■ Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721 123	537						

Second pass

■ Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721 123	537			67			

Second pass

■ Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721 123	537			67	478		

Second pass

■ Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721	537			67	478		
			123	38						

Second pass

■ Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721	537			67	478		
	09		123	38						

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721	537			67	478		
	09		123	38						

- Output of second pass: 3, 9, 721, 123, 537, 38, 67, 478
- used as the input of second pass

Third pass

■ Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's										

■ Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

■ Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003									

■ Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

■ Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009									

■ Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

■ Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009							721		

■ Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

■ Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009	123						721		

■ Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

■ Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009	123				537		721		

■ Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

■ Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003	123				537		721		
	009									
	038									

■ Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

■ Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003	123				537		721		
	009									
	038									
	067									

■ Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

■ Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003	123			478	537		721		
	009									
	038									
	067									

■ Output: 3, 9, 38, 67, 123, 537, 721.

Complexity of Radix sort

- Input size, n
- In our example,
 - Number of buckets, $B = 10$
 - Maximum value, three digits, $M < 10^3$
 - Number of passes, $P = \log_B M = 3$
- Run time proportional to

$$(B + n)P = (B + n) \log_B M \quad (2)$$

- In practice, B should be much larger than 10, so that P is a small number.

- 1 Evaluation of sorting algorithms
- 2 Divide & Conquer. Improvement of merge sort: Quick sort
- 3 Improvement of of insertion sort: Shell Sort
- 4 Beyond comparison based sorting; Bucket and Radix Sort
- 5 Comparison of sorting algorithms

Comparison of sorting algorithms

	Method	Time			In-place	Stable
		Best	Worst	Avg		
Comparison-based	Selection	n^2	n^2	n^2	$O(1)$	No
	Insertion	n	n^2	n^2	$O(1)$	Yes
	Shell			$n^{6/5}$	$O(1)$	No
	HeapSort	$n \log n$	$n \log n$	$n \log n$	$O(1)$	No
	Merge	$n \log n$	$n \log n$	$n \log n$	$O(n)$	Yes
Counting-based	Quick	$n \log n$	n^2	$n \log n$	$O(1)$	No
	Bucket			$O(n+B)$	$O(B)$	Yes
	Radix			$O(n+B)$	$O(B)$	Yes

- Quick sort is often used in practice. (e.g., in Java)
 - It is in-place sorting. Save space compared with mergeSort.
- merge Sort: good for external memory

Takeaways

- Evaluation of sorting algorithms: Time, space, stability, data dependency, application dependency
- Quick sort. Divide and conquer. Time complexity. Worst case and average case. Its comparison with merge sort.
- Shell sort.
- Bucket and Radix sort.
- Readings: Goodrich et al. P532-562.
- [YouTube video for 15 sorting algorithms in 6 minutes](#)